Homework 6

October 23, 2026

Problem 1

Let $\mathscr{A}$ be a linear transformation from a four dimensional linear space $\mathcal{V}$ to itself, $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ be a basis of the linear space $\mathcal{V}$, and the matrix representation of $\mathscr{A}$ with respect to this basis be

$$A = \begin{pmatrix} 2 & 0 & 2 & 1 \\ 6 & 1 & 4 & 4 \\ 10 & 0 & 0 & 4 \\ 7 & 0 & -7 & 2 \end{pmatrix}.$$

Find out

(1) the eigenvalues and eigenvectors of matrix A;

(2) the eigenvalues and associated eigenvectors of $\mathscr{A}$.

Problem 2

Let

$$A = \begin{pmatrix} 4 & 6 & 0 \\ -3 & -5 & 0 \\ -3 & -6 & 1 \end{pmatrix}.$$

Find out the invariant polynomials of A, the natural normal form N of A, and the Jordan canonical forms of A (the complex case and the real case, respectively).

Problem 3

Let A be a matrix whose elementary divisors are

$$(\lambda+1), \quad (\lambda+1)^2, \quad (\lambda+1+j), \quad (\lambda+1-j), \quad (\lambda+1)^2, \quad (\lambda+1+j)^2, \quad (\lambda+1-j)^2.$$

Find out

(1) the Smith canonical form S of the eigenmatrix $\lambda E - A$;

(2) the natural normal form N, the complex Jordan canonical form J_{complex} and the real Jordan canonical form J_{real} of A.